Angles

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 100 minutes · Total marks: 100

KEY RULES

Straight line	Angles on a straight line sum to 180°
Around a point	Sum to 360°
Triangle	Angles sum to 180°; isosceles: two equal angles
Parallel lines	Alternate (Z) equal; corresponding (F) equal; co-interior (C) sum 180°
Polygons	Interior sum $= (n-2) \times 180^\circ$; exterior sum $= 360^\circ$

WORKED EXAMPLE 1

[3 marks]

Angles on a straight line are $(3x + 20)^\circ$ and $(2x + 10)^\circ$. Find x and the two angles.

Step 1	Angles on a straight line add to 180	$3x + 20 + 2x + 10 = 180$
Step 2	Solve	$5x = 150, \quad x = 30$
Step 3	The angles	110° and 70°

Answer: $x = 30$

WORKED EXAMPLE 2

[3 marks]

The angles of a triangle are x°, $2x^\circ$ and $(x + 40)^\circ$. Find the three angles.

Step 1	Angles in a triangle add to 180	$x + 2x + x + 40 = 180$
Step 2	Solve	$4x = 140, \quad x = 35$
Step 3	The angles	$35^\circ, \ 70^\circ, \ 75^\circ$

Answer: $35^\circ, \ 70^\circ, \ 75^\circ$

WORKED EXAMPLE 3

[3 marks]

Work out the size of one interior angle of a regular hexagon.

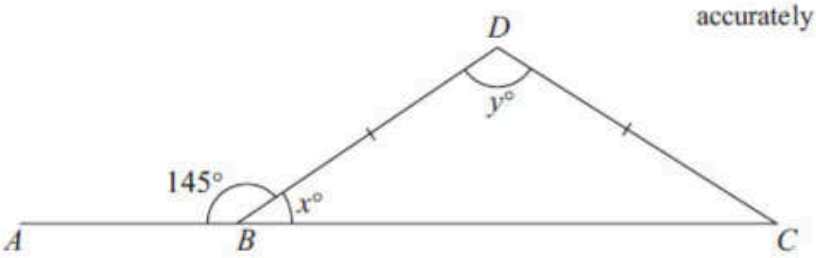
Step 1	Exterior angle = 360 ÷ number of sides	$360 \div 6 = 60^\circ$
Step 2	Interior = 180 − exterior	$180 - 60 = 120^\circ$

Answer: 120°

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1

Diagram **NOT** accurately drawn



ABC is a straight line.
BCD is an isosceles triangle with *DB* = *DC*

Work out the value of

(i) *x*

x = ...

(1)

(ii) *y*

y = ...

(1)

(Total for question = 2 marks)

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Angles

2 The diagram shows a square *ABCD* and a regular pentagon *CDEFG*.

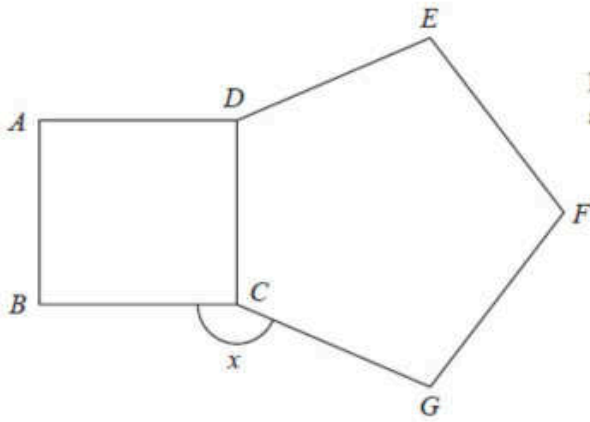


Diagram **NOT** accurately drawn

Work out the size of the angle marked *x*.

..... °

(Total for question = 3 marks)

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3 Each interior angle of a regular polygon is 162°

Work out the number of sides the polygon has.

.....

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4 The diagram shows a pentagon.

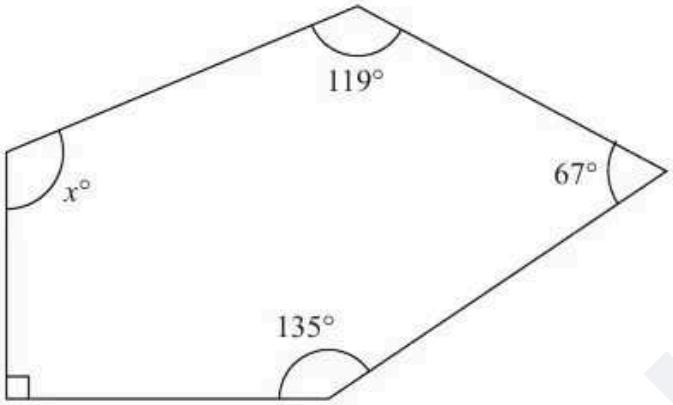


Diagram **NOT** accurately drawn

Work out the value of x

$x =$ .....

(Total for Question 4 is 3 marks)

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5 **6** $ABCDE$ is a pentagon.

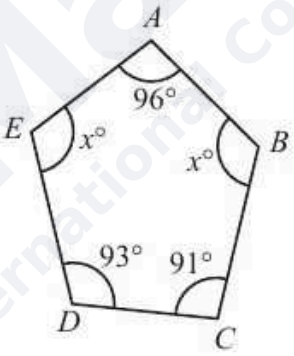


Diagram **NOT** accurately drawn

angle $EAB = 96°$ angle $BCD = 91°$ angle $CDE = 93°$ angle DEA = angle $ABC = x°$

Work out the value of x

$x =$ .....

(Total for Question 6 is 3 marks)

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Angles

6

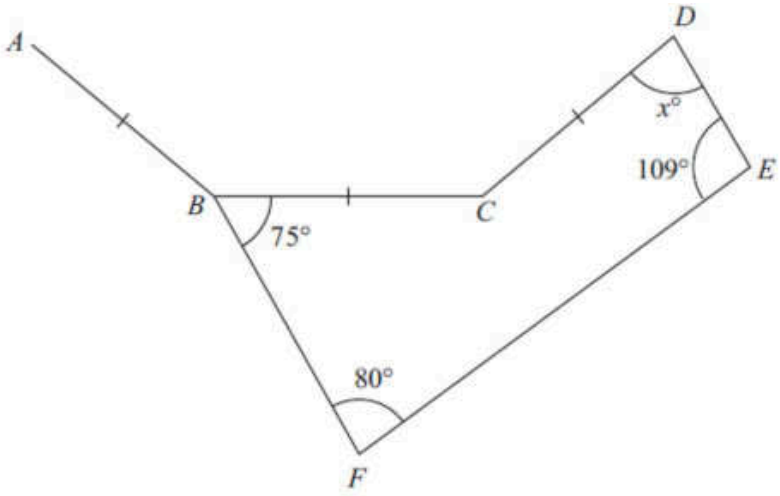


Diagram **NOT** accurately drawn

BCDEF is a pentagon.
AB, *BC* and *CD* are three sides of a regular 10-sided polygon.

Find the value of *x*

x = ...

(Total for question = 3 marks)

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Angles

7 Here is a quadrilateral.

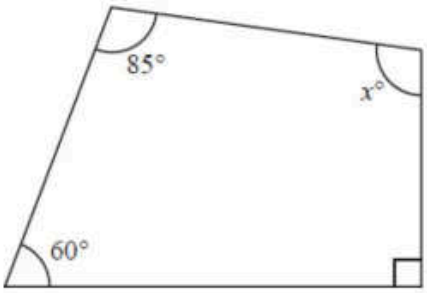


Diagram **NOT** accurately drawn

(a) Work out the value of x

$x =$ .....

(2)

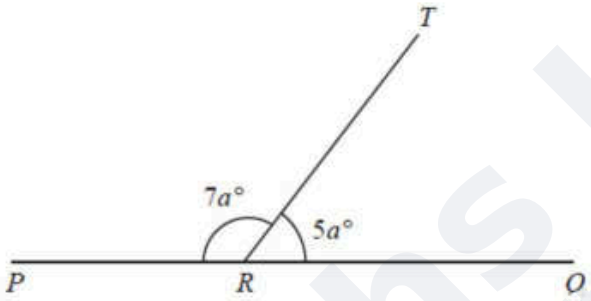


Diagram **NOT** accurately drawn

PRQ is a straight line.

(b) Work out the value of a

$a =$ .....

(2)

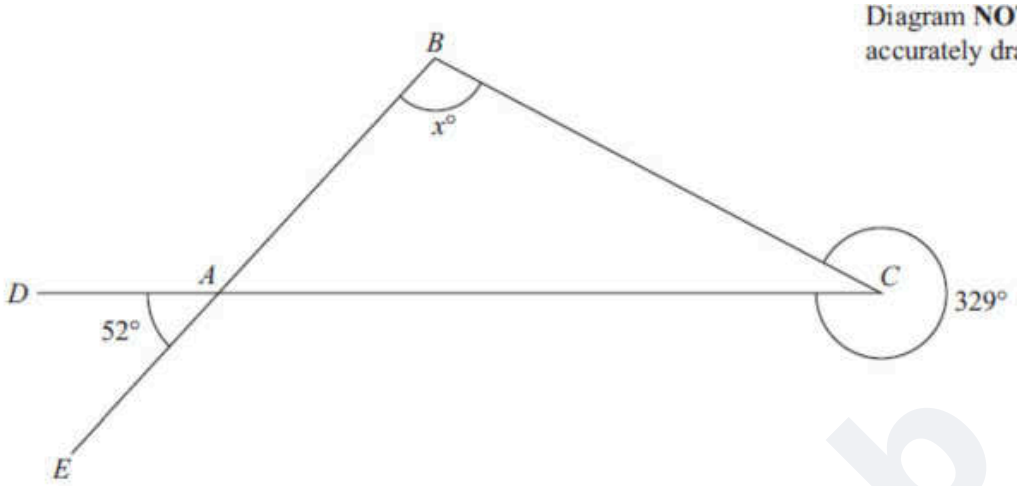
(Total for question = 4 marks)

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8

Diagram **NOT** accurately drawn



ABC is a triangle.
DAC and *EAB* are straight lines.

Work out the value of *x*

x = ...

(Total for question = 4 marks)

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Angles

9

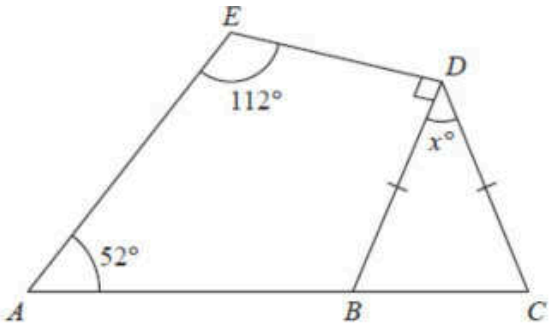


Diagram **NOT** accurately drawn

BCD is an isosceles triangle with *BD* = *CD*
ABC is a straight line.
ABDE is a quadrilateral.

Work out the value of *x*
Give a reason for each stage of your working.

x = ...

(Total for question = 4 marks)

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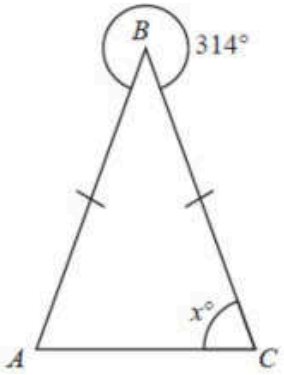


Diagram **NOT** accurately drawn

ABC is an isosceles triangle.

$AB = CB$ reflex angle $ABC = 314°$ angle$ACB = x°$

Work out the value of x
Give a reason for each stage of your working.

$x =$ .....

(Total for question = 4 marks)

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11 *ABCD* is a trapezium.

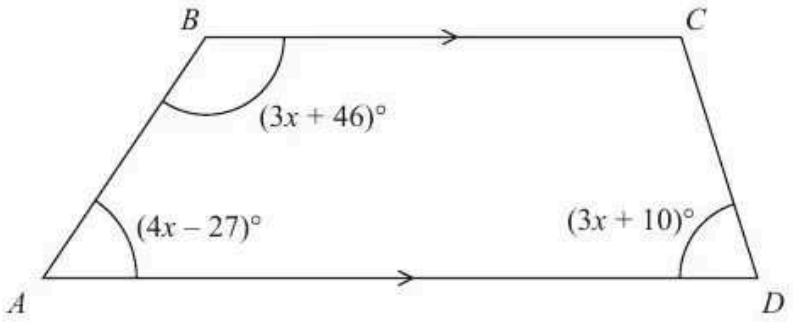


Diagram **NOT** accurately drawn

BC is parallel to *AD*

Find the size of the largest angle inside the trapezium.

.....°

(Total for Question 3 is 4 marks)

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12 **4** The diagram shows a regular octagon $ABCDEFGH$ and a regular pentagon $ABIJK$

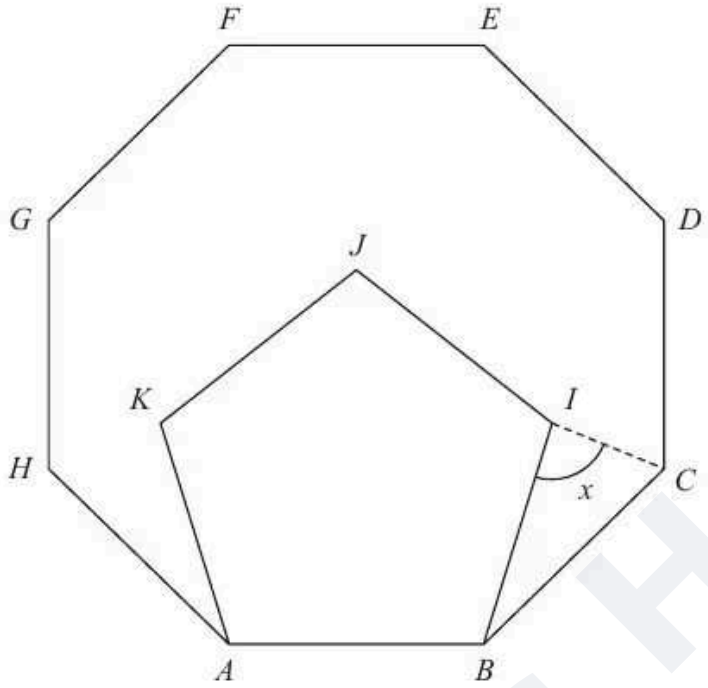


Diagram **NOT** accurately drawn

Work out the size of the angle x

.....

(Total for Question 4 is 4 marks)

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Angles

13 In the diagram, $ABCDE$ is a regular pentagon.

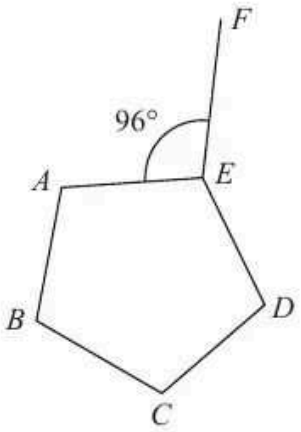


Diagram **NOT** accurately drawn

Angle $AEF = 96°$

Work out the size of the obtuse angle FED
Show your working clearly.

.....°

(Total for Question 5 is 4 marks)

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14 The diagram shows the triangle PQR.

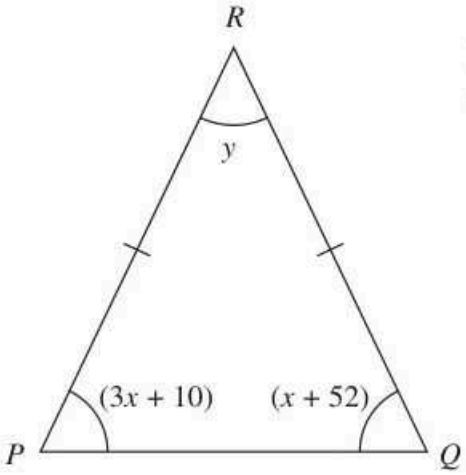


Diagram **NOT** accurately drawn

In the diagram, all the angles are in degrees.

$RP = RQ$

Find the value of y.
Show clear algebraic working.

$y =$

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Angles

15 The diagram shows parts of three regular polygons, **A**, **B** and **C**, meeting at a point.

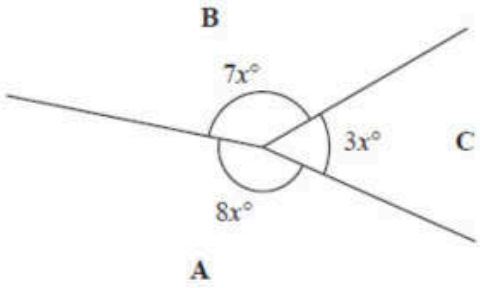


Diagram **NOT** accurately drawn

Polygon **B** has n sides.

Work out the value of n.

n = ...

(Total for question = 4 marks)

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16

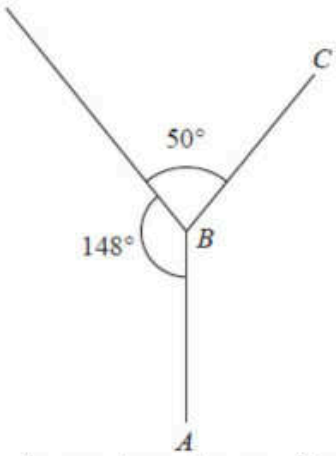


Diagram **NOT** accurately drawn

AB and *BC* are two sides of a regular polygon with *n* sides.

Work out the value of *n*
Show your working clearly.

n = ...

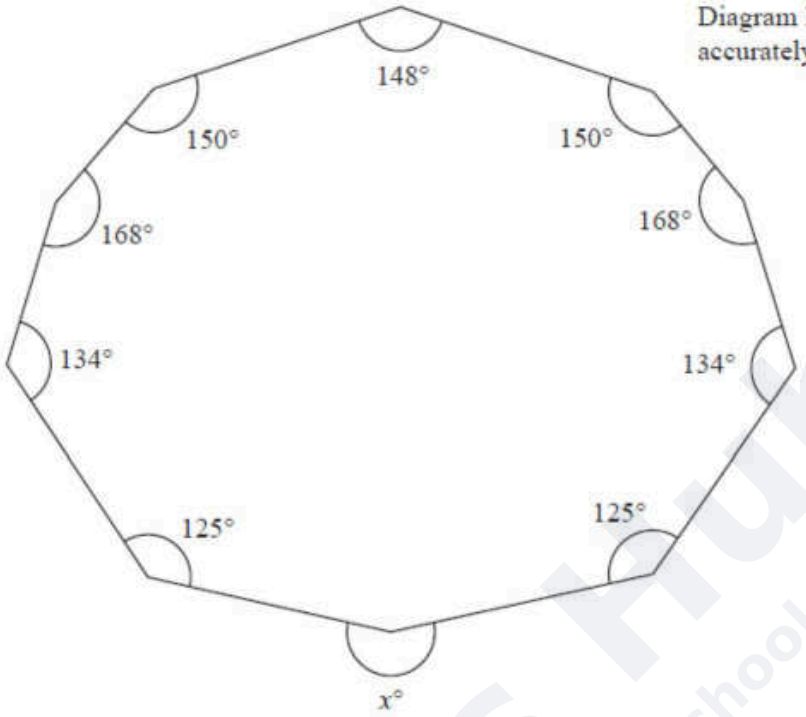
(Total for question = 4 marks)

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Angles

17 Here is a 10-sided polygon.



Work out the value of x.

$x =$ .....

(Total for question = 4 marks)

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18

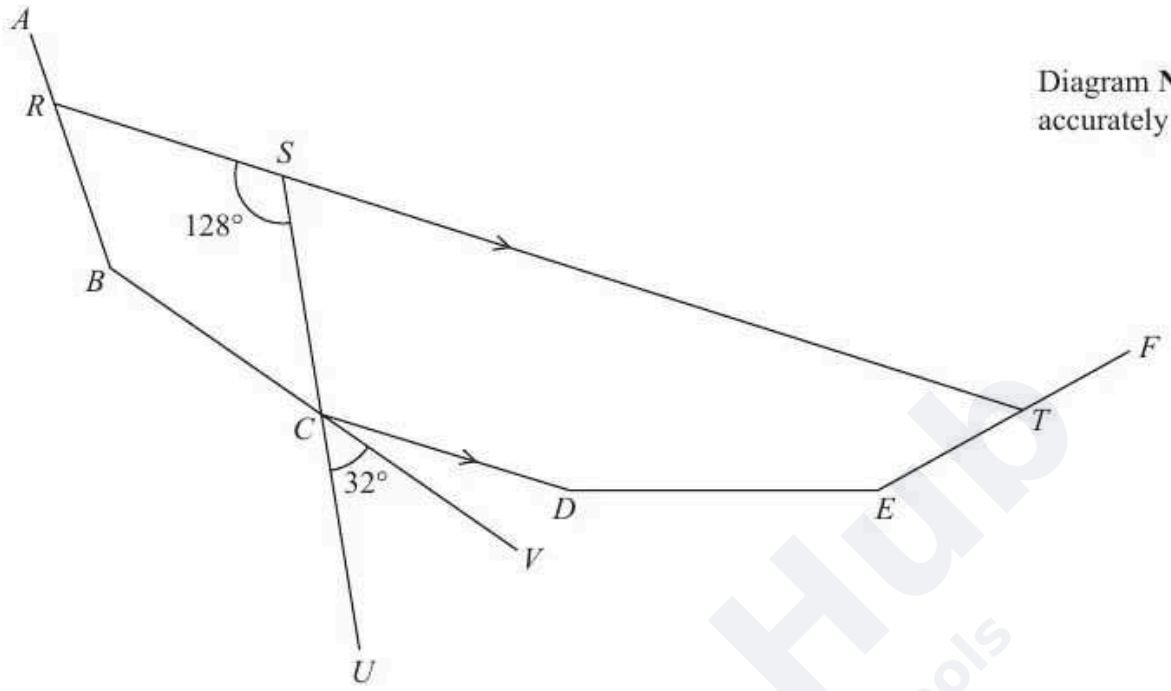


Diagram **NOT** accurately drawn

B, *BC*, *CD*, *DE* and *EF* are five sides of a regular polygon.

ST, *SCU* and *BCV* are straight lines.
ST is parallel to *CD*
ngle *RSC* = 128°
ngle *UCV* = 32°

Work out how many sides the polygon has.
how your working clearly.

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19

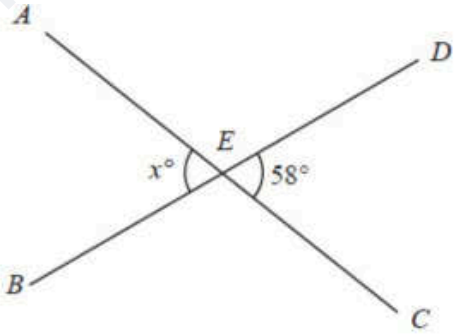


Diagram **NOT** accurately drawn

AEC and *BED* are straight lines.

Sergio says that the value of x is 58

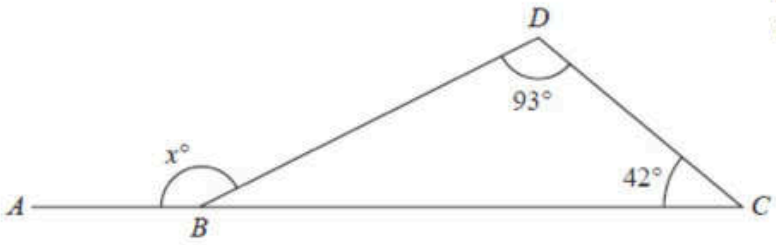
(a) Give a reason why Sergio is correct.

.....

(1)

20 *ABC* is a straight line and *BCD* is a triangle.

Diagram **NOT** accurately drawn



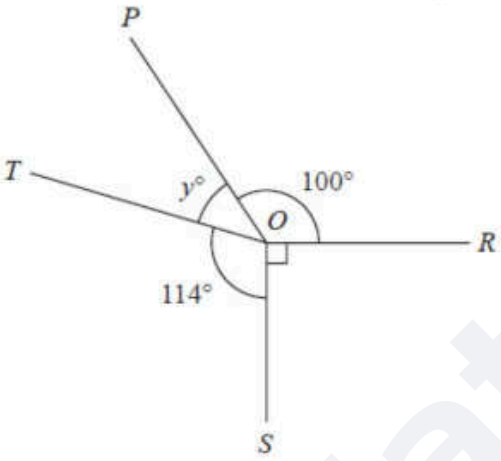
(a) Work out the value of *x*

x = ...

(2)

PO, *RO*, *SO* and *TO* are four straight lines.

Diagram **NOT** accurately drawn



(b) (i) Work out the value of *y*

y = ...

(2)

(ii) Give a reason for your answer.

.....

(1)

(Total for question = 5 marks)

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CHALLENGE QUESTIONS

21

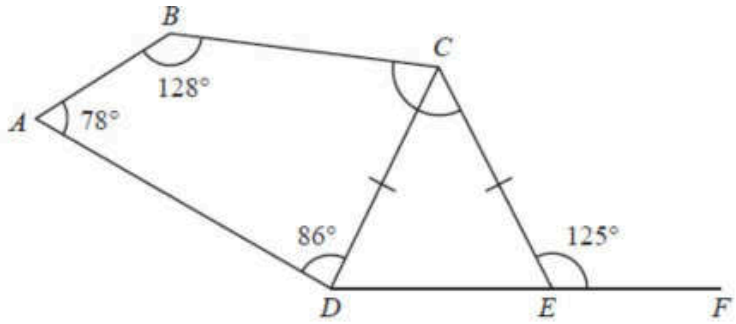


Diagram **NOT** accurately drawn

ABCD is a quadrilateral.
CDE is an isosceles triangle with *CD* = *CE*
DEF is a straight line.
Work out the size of angle *BCE*
Give a reason for each stage of your working.

angle *BCE* = ... °

(Total for question = 5 marks)

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Angles

22 The diagram shows a regular octagon *ABCDHIJK* and a pentagon *DEFGH*.

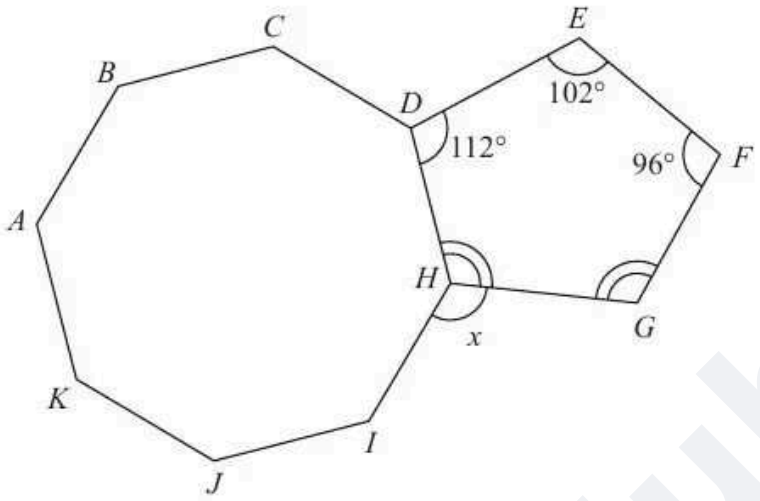


Diagram **NOT** accurately drawn

Angle *GHD* = angle *FGH*.

Work out the size of the angle marked *x*.
Show your working clearly.

.....°

(Total for Question 5 is 5 marks)

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23 5

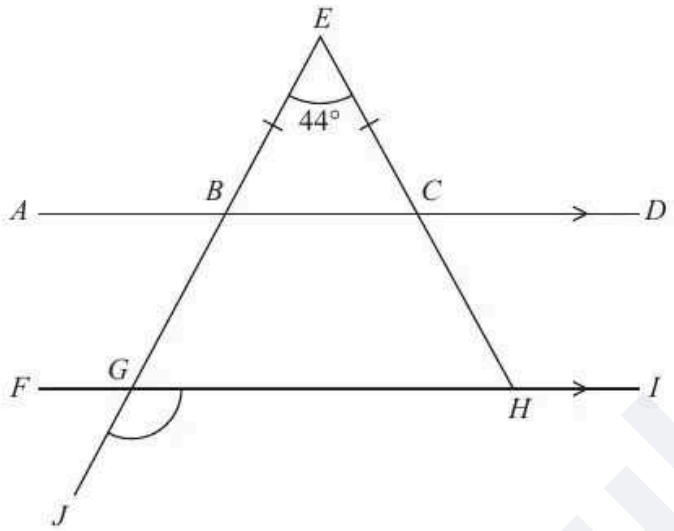


Diagram **NOT** accurately drawn

ABCD and *FGHI* are parallel straight lines.
EBGJ and *ECH* are straight lines.

BE = *CE*
Angle *BEC* = 44°

Work out the size of angle *JGH*.
Give a reason for each stage of your working.

.....°

(Total for Question 5 is 5 marks)

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24 The diagram shows a regular pentagon, *ABCDE*, a regular hexagon, *CFGHID*, and a quadrilateral, *EDIJ*.

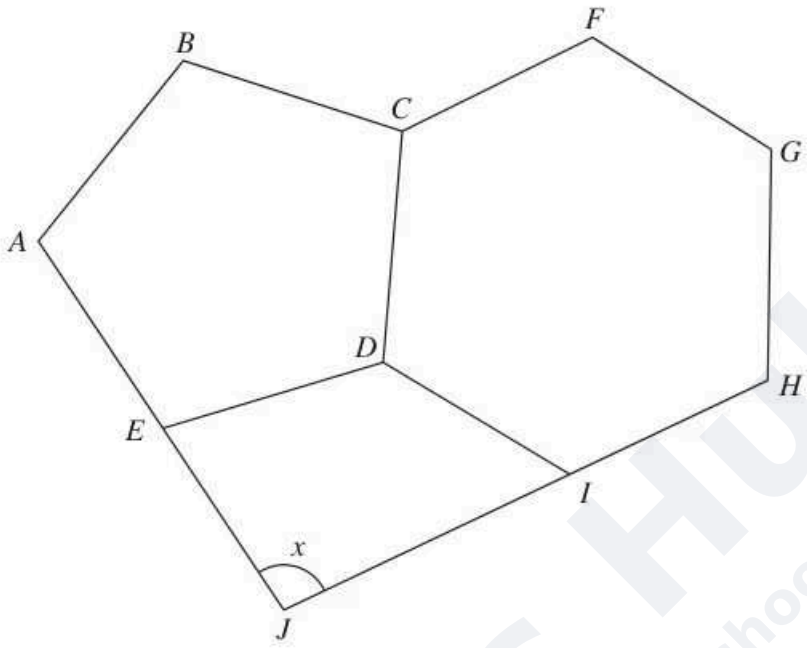


Diagram **NOT** accurately drawn

AEJ and *HIJ* are straight lines.

Work out the size of the angle marked *x*.
Show your working clearly.

.....°

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25 **11**

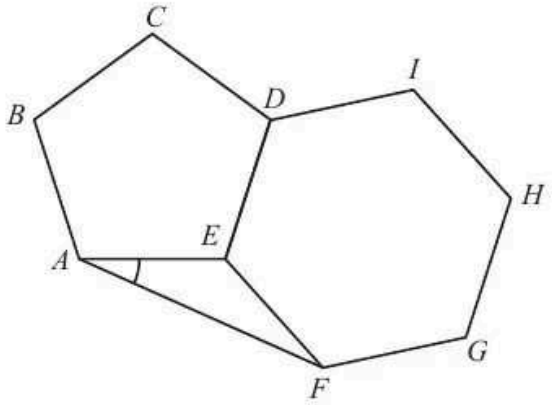


Diagram **NOT** accurately drawn

ABCDE is a regular pentagon.
DEFGHI is a regular hexagon.

AF is a straight line.

Work out the size of angle *EAF*

.....

(Total for Question 11 is 5 marks)

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TOTAL FOR PAPER IS 100 MARKS

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